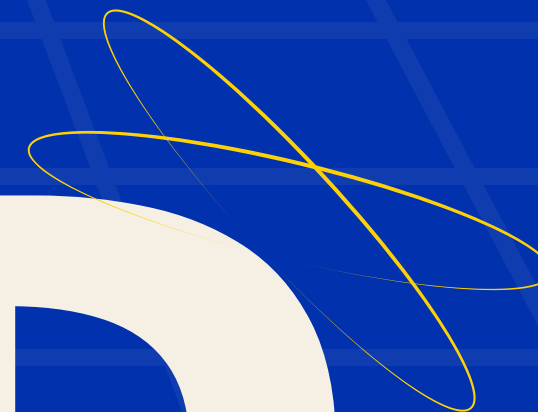
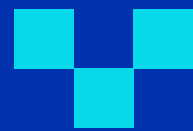




ROUND TABLE



— PROJECT PROPOSAL —

Presented by : The Keystrokes

October 4, 2026

THE

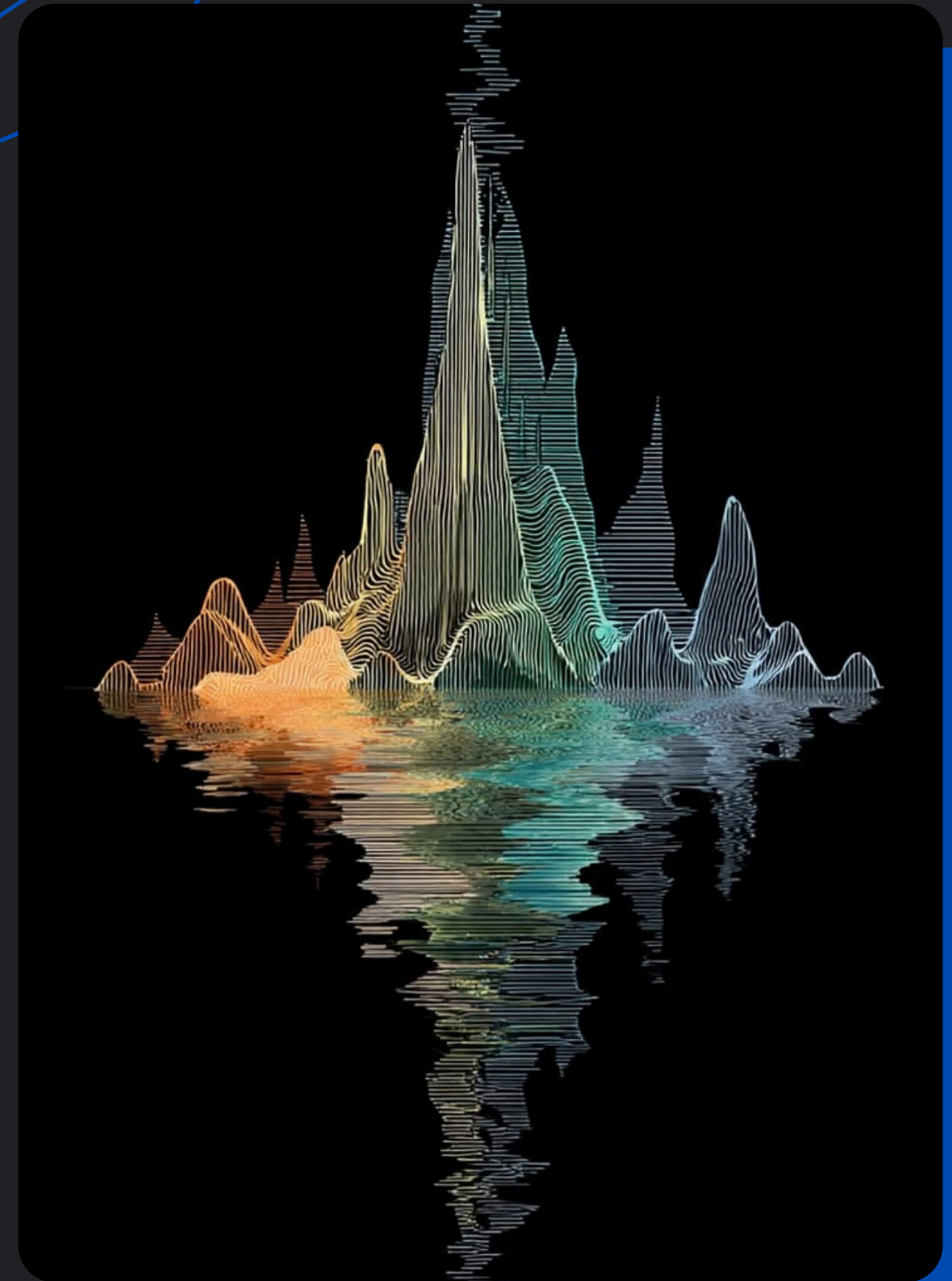
BREAKDOWN

The Cocktail Party Problem

In group meetings, people naturally overlap and interrupt. Monolithic ASR (like standard Zoom or YouTube captions) collapses multi channel speech into a single garbled stream, dropping accuracy by over 40% during cross-talk.

Cognitive Overload & Exclusion

DHH (Deaf and Hard accuracy by over 40% during cross-talk. Header: Cognitive Overload & of Hearing) participants cannot identify who spoke what without visual speaker attribution. Current caption tools flicker constantly and lag by 2-4 seconds, making real-time participation impossible.



PIPELINE

LOW LATENCY STREAMING ARCHITECTURE

1. Mic Audio

2. VAD

3. Speaker
Diarization



1. Ingestion & VAD

WebRTC captures raw audio chunks. Silero VAD filters background room noise and detects active voice frames in under 30ms.



2. Streaming Diarization

pyannote-based online voiceprint embeddings cluster unique speaker profiles on the fly without waiting for audio segments to finish.

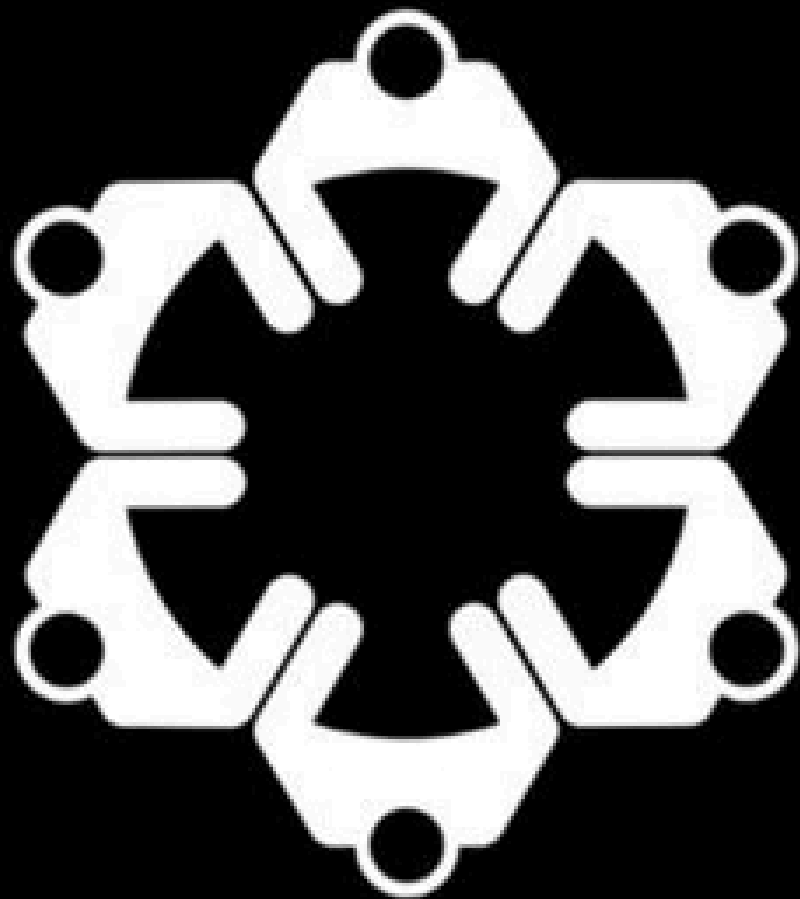


3. Real-Time ASR

Quantized Whisper (CTranslate2/ONNX) outputs interim token streams with <200ms glass-to-glass latency, separating simultaneous speech turns.

USER EXPERIENCE

SOLVING CROSS-TALK WITH PARALLEL VISUAL STREAMS



1. Dynamic Parallel Split

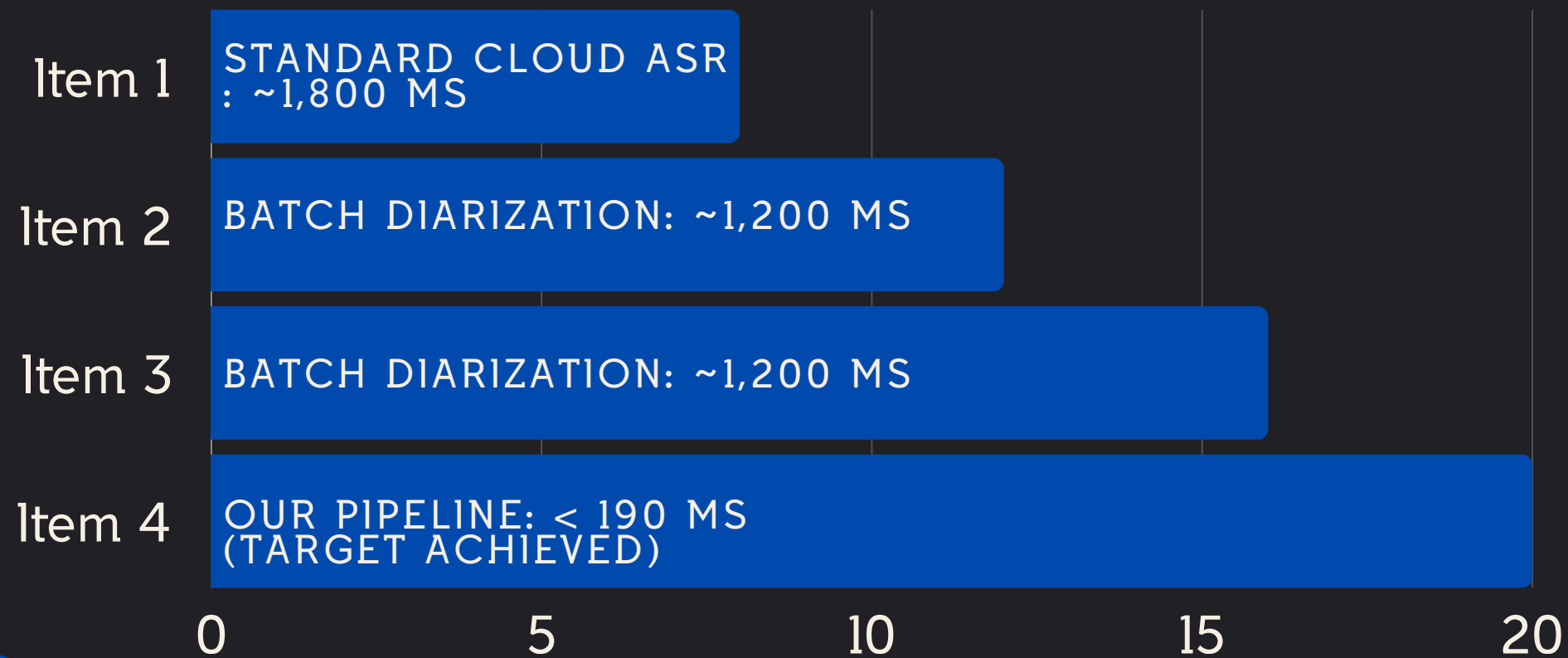
When two people talk simultaneously, the caption dock dynamically bifurcates into two independent side-by-side cards. This prevents text collisions and lets the reader follow both speakers naturally.

2. Dual-Coded Attribution & Visual Stability

Combines accessible color palettes with avatar badges for zero-guess speaker identification. Live streaming words are rendered in muted italics, locking into solid contrast once finalized to eliminate reading flicker.

BENCHMARKS

MEASURED LATENCY, ACCURACY, AND EFFICIENCY



< 7.5%

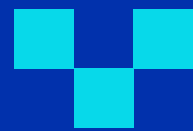
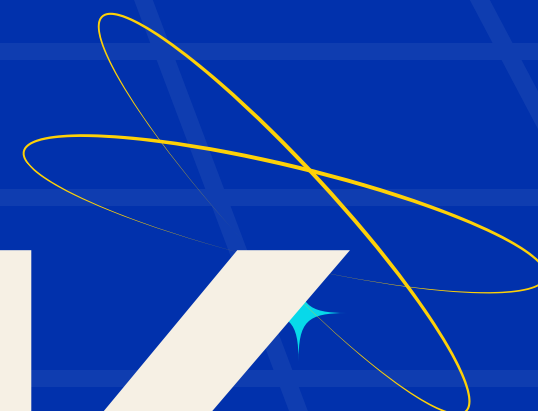
Diarization Error Rate (DER)

Tested against overlapping conversational datasets (AMI Corpus standard).

0.00 ₹

Cloud Server Cost (Local Edge Engine)

Runs client-side via WebAssembly/ONNX for complete conversational privacy



THANK YOU

Presented by :
The Keystrokes | Hackathon '26

Oct 04, 2026